Jacobian-Free Krylov-Arnoldi Framework for Static Voltage Stability Estimation of Power Systems With 100% Renewable Energy

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***Abstract*—The pervasive adoption of inverter-interfaced resources in 100% renewable power systems fundamentally alters voltage regulation dynamics and renders classical margin-estimation techniques computationally prohibitive. This paper introduces a unified, Jacobian-free Krylov-Arnoldi (JFKA) framework for automated estimation of system static voltage stability (SVS). First, inverter current-limiting behavior is captured by a smooth S-type function, preserving continuous power-flow structure across both grid-forming (GFM) and grid-following (GFL) control modes. Building on singularity theory, we derive a Jacobian-free stability indicator that pinpoints the onset of voltage collapse without explicit derivative evaluation. To efficiently solve the resulting large-scale, transcendental power-flow equations, we embed a reduced-order Arnoldi process within a Newton-Krylov solver, yielding rapid convergence and markedly lower memory footprint. Case studies on a modified IEEE-39 bus network with full renewable penetration demonstrate that our method accurately tracks voltage regulation limits under varied load-growth scenarios and automatic mode switches.**

***Note to Practitioners*—As grids reach 100% renewable energy penetration, real-time voltage regulation capability estimation must handle inverter mode switching and large-scale nonlinear solves. This work introduces a JFKA solver that uses an S-type function for smooth current-limit modeling and a singularity-based indicator for collapse detection. Integrated into existing EMS/DA tools, it enables continuous margin tracking under load growth and mode shifts. The framework runs on standard desktop hardware and scales to large power grids, offering plug-and-play deployment for automated voltage regulation capability assessment.**

***Index Terms*—Power systems with 100% renewable energy, voltage regulation capability estimation, renewable energy sources, Jacobian-free Krylov-Arnoldi framework.**

I. Introduction

THE urgent need to mitigate climate change has placed the decarbonization of electrical power systems at the forefront of global energy policy. As nations commit to increasingly ambitious zero-carbon targets, the transition to power systems comprised entirely of renewable energy sources (REPSs) is accelerating worldwide [1], [2]. In these 100% REPSs, conventional synchronous generators (SGs) are being displaced by inverter-interfaced wind and solar installations, resulting in fundamental shifts in both operational paradigms and system dynamics [3]. Numerous regions—including major economies—have pledged to achieve full renewable supply by 2050, underscoring the urgency and scale of this transformation [4], [5], [6].

Unlike SGs, inverter-based resources regulate power flow by modulating DC voltages through power electronic switching. This introduces stringent current limits, necessary for protecting sensitive semiconductor devices from thermal and electrical stress [9]. As grid conditions evolve—such as fluctuating bus voltages and rising demand—inverters must dynamically switch between grid-forming (GFM) and grid-following (GFL) modes. During periods of stress, current-limiting action further constrains their ability to provide reactive power support. These nonlinear and mode-dependent behaviors present new technical challenges for system operators, particularly in the context of static voltage stability (SVS) estimation across large-scale, highly renewable grids.

Historically, SVS estimation has relied on classical methodologies such as continuation power flow (CPF) and voltage-sensitivity analysis, which utilize explicit Jacobian matrix calculations and track system stability margins via model-based rollbacks at detected bifurcation points [10], [11]. While advanced methods—including holomorphic embedding [12] and load-impedance modulus margin techniques [13]—have improved computational efficiency in SG-dominated environments, they are inherently limited. These approaches often assume idealized voltage sources, and lack the flexibility to accommodate the continuous, nonlinear control-mode transitions and current-limiting effects intrinsic to inverter-based grids. As a result, they are increasingly ill-suited for real-time monitoring and margin estimation in future REPSs.

Recent research efforts have begun to address the uncertainty inherent in renewable generation, employing Monte Carlo simulations [14], global sensitivity analysis [15], matrix factorization [16], and advanced weather-informed risk assessments [17]. However, these studies predominantly focus on stochastic variability in generation, rather than the structural and dynamical changes in voltage regulation mechanisms

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introduced by inverter control. The evolving landscape of inverter controls—spanning GFM and GFL paradigms, with subcategories such as PQ and PV modes [18], [19]—has a profound impact on system voltage support, particularly under operational stress or rapid switching. While some recent works extend Jacobian formulations [21] or incorporate voltage stability constraints in unit commitment [22], many continue to treat inverter control modes as static, neglecting the dynamic transitions induced by real-time current limiting [23]. This oversimplification can introduce significant errors, especially when estimating SVS near operational boundaries.

Current strategies to model control-mode switching often rely on discrete rollback mechanisms: transition points are detected, the power-flow model is updated, and analysis resumes [24]. While feasible for small networks, this approach becomes computationally intractable in large-scale systems due to the combinatorial growth of model variants and the potential for algorithmic instability. More specifically, CPF methods [10], [11] suffer from high computational cost due to repeated Jacobian evaluations and step size control, making them less suitable for large-scale or real-time applications. Holomorphic embedding techniques [12] may encounter convergence and truncation errors in highly nonlinear systems with extensive inverter penetration. Approaches relying on static inverter models [23] cannot capture the dynamic mode-switching and current-limiting effects of modern inverters, leading to inaccurate SVS estimation under stress or rapid operating changes. Consequently, two major challenges remain unresolved: (1) integrating a continuous representation of mode-switching behavior into SVS estimation without resorting to discrete model updates, and (2) developing a scalable, Jacobian-free computational solver capable of real-time application in large, complex power networks.

Against this backdrop, this paper presents a unified Jacobian-Free Krylov-Arnoldi (JFKA) framework for automated voltage regulation capability estimation in 100% renewable power systems. Our approach is distinguished by the following key innovations:

1) **Jacobian-free collapse indicator:** By leveraging singularity theory, we develop a robust stability indicator that detects proximity to voltage collapse using trends in the minimum singular value of Jacobian-vector products. This obviates the need for explicit derivative calculations, dramatically reducing computational overhead.
2) **Newton-Krylov solver with reduced-order Arnoldi process:** We embed a low-dimensional Arnoldi process within each Newton iteration, enabling efficient and memory-frugal solution of the large-scale and transcendental power-flow equations characteristic of REPSs.
3) **Continuous mode-switching modeling:** The proposed framework captures the effect of control mode transitions—induced by inverter current limiting—on SVS estimation, ensuring accuracy and robustness even under rapid and large-scale operational changes.

The remainder of this paper is organized as follows. Section II explores the impact of inverter control mode switching on SVS. Section III introduces the proposed JFKA method. Section IV presents comprehensive case studies on a modified IEEE-39 bus system with full renewable penetration. Finally, Section V concludes with a summary of key findings and future research directions.

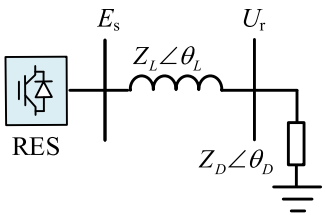


Fig. 1. Single-RES system.

II. Effect of Mode Switching on SVS

n the conventional PV curve analysis, the SG is regarded as a constant voltage source. Since the operating mode of RES will switch as the bus voltage decreases, the analysis results based on the conventional PV curve cannot be applied to RES. To this end, we establish the system model of the single-RES system as shown in Fig. 1.

In Fig. 1, E_s denotes the voltage magnitude of the RES, U_r denotes the voltage magnitude of the load bus, Z_L and θ_L denote the magnitude and phase of the line impedance respectively, Z_D and θ_D denote the magnitude and phase of the load respectively.

When the bus voltage of the RES is kept constant, the load power can be written as a function of system parameters and bus voltage:

$$\begin{aligned} P_D + jQ_D &= \dot{U}_r \bar{\dot{I}}_L = \dot{U}_r \frac{E_s - \bar{\dot{U}}_r}{Z_L\angle -\theta_L} \\ &= \frac{U_r E_s \cos\theta_r - U_r^2 + jU_r E_s \sin\theta_r}{Z_L(\cos\theta_L - j\sin\theta_L)} \end{aligned} \tag{1}$$

where, P_D and Q_D denote the active power and reactive power of the load respectively, $\dot{U}_r$ and $\bar{\dot{U}}_r$ denote the voltage vector of the load bus and its conjugate respectively, $\dot{I}_L$ and $\bar{\dot{I}}_L$ denote the output current vector of the RES and its conjugate respectively.

Neglecting the line resistance in (1), we can get:

$$\begin{cases} U_r \sin\theta_r = -\dfrac{P_D X_L}{E_s} \\ U_r \cos\theta_r = \dfrac{1}{E_s} U_r^2 + \dfrac{Q_D X_L}{E_s} \end{cases} \tag{2}$$

Eliminating the variable θ_r in (2) yields:

$$U_r^4 + \left(2Q_D X_L - E_s^2\right) U_r^2 + X_L^2\left(P_D^2 + Q_D^2\right) = 0 \tag{3}$$

Two solutions of the bus voltage can be obtained by solving (3):

$$U_r = \sqrt{\frac{1}{2}E_s^2 - Q_D X_L \pm \frac{1}{2}\sqrt{\left(2Q_D X_L - E_s^2\right)^2 - 4X_L^2\left(P_D^2 + Q_D^2\right)}} \tag{4}$$

In (4), the voltage magnitude of the load bus is obtained under the assumption that the RES keeps the terminal voltage constant. If the RES is switched to constant current control

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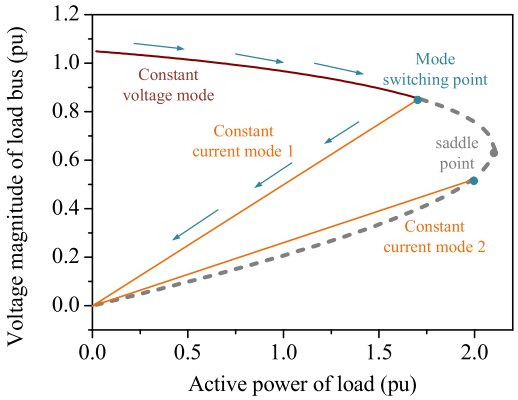


Fig. 2. Schematic diagram of PV curve considering RES current limit.

TABLE I
THE SYSTEM PARAMETERS

Items	Values
Terminal voltage of the RES	1.05
Power factor of the load	0.95
Impedance of the line	0.2
Capacity of RES 1	2.0
Capacity of RES 2	5.0

mode, the output current magnitude of the RES will remain constant, then the load bus voltage is:

$$U_{\mathrm{r}} = \frac{1}{I_{\mathrm{lim}}} \sqrt{P_{\mathrm{D}}^2 + Q_{\mathrm{D}}^2} \tag{5}$$

where, I_{lim} denote the current limit.

Assuming that the power factor of the load keeps unchanged, then the voltage magnitude of the load bus is proportional to the active load. In the PV plane, the power-voltage characteristic of the RES is represented by a straight line crossing the origin. The PV curve considering the RES current limitation is shown in Fig. 2, and the system parameters are presented in Table I.

As the load increases from 0, the voltage magnitude of the load bus will decrease from the initial value E_{s}. When the voltage of the load bus decreases to a small value, the required current will exceed the limit of the RES, which will cause the RES to change from constant voltage control mode to constant current control mode. If the capacity of RES is not large enough, (set to 2.0 pu in this case), then the RES will switch before the operating point reaches the saddle point. In Fig. 2, when the operating point reaches the mode switching point, as the load impedance decreases, the operating point will shift from the PV curve in constant voltage control mode to the PV curve in constant current control mode. If the load impedance continues to decrease, the system operating point will move toward the origin along the PV curve in constant current control mode. In this case, the load limit of the system is not at the saddle point, but at the mode switching point. If the capacity of the RES is large enough (set to 5.0 pu in this case), then the PV curve in constant current control mode will be below the saddle point. As the load impedance decreases, the operating point of the system will first reach the saddle point, and then reach the control mode switching point after crossing the saddle point. In this case, the load limit of the system is still at the saddle point.

III. JFKA Framework for SVS Estimation

A. Generalized Power Flow Model of 100% REPSs Considering Mode Switching

The generalized power flow model considering mode switching is proposed for 100% REPSs, which can ensure the form invariance of the system model. The tradition power equation is adopted to describe the load and connection buses:

$$\begin{cases} P_{si} = \sum_{j=1}^{n} U_i U_j (G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}) \\ Q_{si} = \sum_{j=1}^{n} U_i U_j (G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}) \end{cases} \tag{6}$$

where, P_{si} and Q_{si} denote the injected active and reactive power of bus i, U_i denotes the voltage magnitude of bus i, G_{ij} and B_{ij} denote the real and imaginary part of the element in row i and column j in the admittance matrix respectively, θ_{ij} denotes the phase difference between bus i and bus j.

Depending on the control mode selection of the RES, the buses with RES connected can also be divided into three types: PQ bus, PV bus and slack bus. Different from the SG, RES is very sensitive to the current magnitude and has a small value for current limiting, making it easy to reach the current limit. For a bus connected with RES, its current balance equation is:

$$\begin{cases} I_{\mathrm{res}i} \cos\delta_i = \sum_{j\in i} \left(G_{ij} U_j \cos\theta_j - B_{ij} U_j \sin\theta_j\right) \\ I_{\mathrm{res}i} \sin\delta_i = \sum_{j\in i} \left(B_{ij} U_j \cos\theta_j + G_{ij} U_j \sin\theta_j\right) \end{cases} \tag{7}$$

where, $I_{\mathrm{res}i}$ denotes the injected current of RESi, $j \in i$denotes that bus j is directly connected to bus i, δ_i denotes the phase angle of RES output current referenced to the system reference bus, θ_i denotes the phase of bus i.

To ensure the form invariance of the system model, a continuous S-type function is used to fit the effect of the limiting link:

$$I_{\mathrm{res}i} = I_{\mathrm{lim}\,i} \left[\frac{2}{1 + \exp\left(\sum_{j=1}^{k} -\alpha_j \left(\frac{I_{\mathrm{ref}i}}{I_{\mathrm{lim}\,i}} \right)^{2j-1} \right)} - 1 \right] \tag{8}$$

where, $I_{\mathrm{lim}\,i}$ denotes the current limit of RESi, $I_{\mathrm{res}i}$ denotes the output current magnitude of RES i, α_j denotes the coefficient, $I_{\mathrm{ref}i}$ denotes the reference of the output current of RES i.

According to (8), the highest power of the natural constant in the denominator is $2k$-1, and the fitting effect will be different when choosing different k. Fig. 3 shows the fitting results with different powers. It can be seen from Fig. 3 that when k reaches 3, the error between the S-type function and the limiting link can be almost ignored. Therefore, the S-type function can simultaneously guarantee the calculation accuracy and the form invariance of the system model.

The input of the S-type function is current reference I_{ref}, and the output is actual current I_{res}. The current reference is jointly

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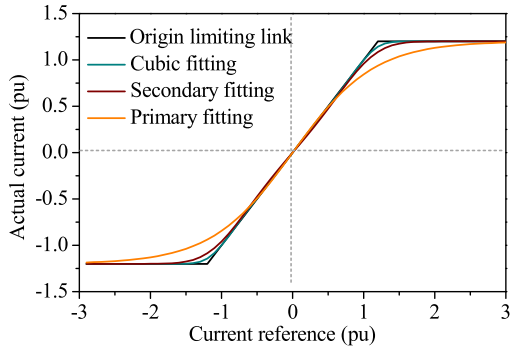


Fig. 3. Fitting results of the S-type function with different powers.

determined by the terminal voltage and the control mode of RES. When RES operates in PQ control mode, its expected active power and reactive power are given by operators, and the current reference of RES can be obtained by solving the following equations:

$$\begin{cases} P_{\text{refi}} = P_{Li} + U_i \cos\theta_i I_{\text{refi}} \cos\delta_i + U_i \sin\theta_i I_{\text{refi}} \sin\delta_i \\ Q_{\text{refi}} = Q_{Li} + U_i \cos\theta_i I_{\text{refi}} \sin\delta_i - U_i \sin\theta_i I_{\text{refi}} \cos\delta_i \end{cases} \tag{9}$$

where, P_{refi} and Q_{refi} denote the active/reactive power reference of RES respectively.

When RES operates in PV control mode, its expected active power and bus voltage are given by operators, the current reference of RES can be obtained by solving the following equations:

$$\begin{cases} P_{refi} = U_{refi} \cos\theta_i' I_{refi} \cos\delta_i + U_{refi} \sin\theta_i' I_{refi} \sin\delta_i \\ I_{refi} \cos\delta_i = \dfrac{P_{Li}\cos\theta_i + Q_{Li}\sin\theta_i}{U_i} + \\ \sum_{j\neq i} \left(G_{ij}U_j \cos\theta_j - B_{ij}U_j \sin\theta_j\right) + G_{ii}U_{refi}\cos\theta_i' \\ \qquad -B_{ii}U_{refi}\sin\theta_i' \\ I_{refi} \sin\delta_i = \dfrac{P_{Li}\sin\theta_i - Q_{Li}\cos\theta_i}{U_i} + \\ \sum_{j\neq i} \left(B_{ij}U_j \cos\theta_j + G_{ij}U_j \sin\theta_j\right) + B_{ii}U_{refi}\cos\theta_i' \\ \qquad +G_{ii}U_{refi}\sin\theta_i' \end{cases} \tag{10}$$

where, θ_i' denotes the phase reference of RES under the expected active power and terminal voltage, which is the quantity to be sought, and U_{refi} denotes the expected terminal voltage magnitude of RES.

When the RES operates in a slack control mode, its expected terminal voltage magnitude and phase are given by operators, the current reference of RES can be obtained by solving the following equations:

$$\begin{cases} I_{\text{refi}} \cos\delta_i = \sum_{j\neq i} \left(G_{ij}U_j \cos\theta_j - B_{ij}U_j \sin\theta_j\right) + \\ G_{ii}U_{\text{refi}} \cos\theta_{\text{refi}} - B_{ii}U_{\text{refi}} \sin\theta_{\text{refi}} \\ I_{\text{refi}} \sin\delta_i = \sum_{j\neq i} \left(B_{ij}U_j \cos\theta_j + G_{ij}U_j \sin\theta_j\right) + \\ B_{ii}U_{\text{refi}} \cos\theta_{\text{refi}} + G_{ii}U_{\text{refi}} \sin\theta_{\text{refi}} \end{cases} \tag{11}$$

where, θ_{refi} is the expected terminal voltage phase of RES. In this paper, a phase angle variable θ_i is assigned to each bus for notational consistency. The slack bus is selected as the reference, and its phase angle θ_i is fixed at zero, while all other bus phase angles are defined relative to this reference. This constraint is applied in all simulations and solution procedures.

Eq. (6)–(11) constitute the generalized power flow model of the 100% REPS considering the current limit. In the traditional power flow model, the current magnitude of the RES has not been taken into consideration. In the proposed generalized power flow model, the RES current is added as a quantity to be solved, and the influence of current limiting is considered. Since the continuous S-type function is adopted to replace the original limiting link, the form invariance of the system model in the SVS estimation is guaranteed.

B. Jacobian-Free Voltage Stability Criterion

The singularity of the Jacobian matrix is often utilized as a criterion for judging the voltage stability of the system. If the Jacobian matrix is singular, it indicates that the power flow equation has no solution, which physically manifests as voltage collapse. The research on continuous power flow has proved that the singular point of the Jacobian matrix is consistent with the saddle point of the PV curve. In traditional power systems dominated by SGs, the form of the power flow equation is relatively simple, making it easy to derive the Jacobian matrix. In 100% REPSs, the S-type function brings great difficulties to generating the Jacobian matrix. To solve this problem, a Jacobian-free voltage stability criterion is proposed below.

Assuming the Jacobian matrix of the 100% REPS is $\boldsymbol{J}$, the product of $\boldsymbol{J}$ and any vector can be written as:

$$\boldsymbol{J}\boldsymbol{q} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\boldsymbol{q} = \lim_{\Delta x \to 0} \frac{\boldsymbol{f}(\boldsymbol{x}+\Delta\boldsymbol{x}) - \boldsymbol{f}(\boldsymbol{x})}{\Delta \boldsymbol{x}}\boldsymbol{q} \tag{12}$$

where, $\boldsymbol{f}$ denotes the power flow equation set, $\boldsymbol{x}$ denotes the power flow quantity vector, $\boldsymbol{q}$ denotes any given vector.

If we set $\Delta\boldsymbol{x}$ as:

$$\Delta\boldsymbol{x} = \varepsilon\boldsymbol{q} \tag{13}$$

where, ε denotes a small constant.

Substituting (13) into (12), we can get:

$$\boldsymbol{J}\boldsymbol{q} = \lim_{\varepsilon \to 0} \frac{\boldsymbol{f}(\boldsymbol{x}+\varepsilon\boldsymbol{q}) - \boldsymbol{f}(\boldsymbol{x})}{\varepsilon} \tag{14}$$

If we choose an identity orthogonal matrix $\boldsymbol{E}$=[$\boldsymbol{e}_1$, $\boldsymbol{e}_2$, …, $\boldsymbol{e}_n$], according to (14), the product of matrix $\boldsymbol{J}$ and matrix $\boldsymbol{E}$ is:

$$\boldsymbol{J}\boldsymbol{E} = [\boldsymbol{J}\boldsymbol{e}_1, \boldsymbol{J}\boldsymbol{e}_2, \cdots, \boldsymbol{J}\boldsymbol{e}_n] \tag{15}$$

The expression of any column is:

$$\boldsymbol{J}\boldsymbol{e}_i = \mathrm{im}_{\varepsilon \to 0} \frac{\boldsymbol{f}(\boldsymbol{x}+\varepsilon\boldsymbol{e}_i) - \boldsymbol{f}(\boldsymbol{x})}{\varepsilon} \tag{16}$$

According to the properties of the identity orthogonal matrix, matrix $\boldsymbol{J}$ multiplied by matrix $\boldsymbol{E}$ is equivalent to rotating matrix $\boldsymbol{J}$ with the same length. The singular values of the matrix $\boldsymbol{JE}$ are exactly the same as those of matrix $\boldsymbol{J}$. Therefore, we only need to perform singular value decomposition on matrix $\boldsymbol{JE}$ to obtain the minimum singular value of matrix

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$\boldsymbol{J}$. According to equation (16), obtaining matrix $\boldsymbol{JE}$ does not require deriving the expression of the Jacobian matrix, which greatly reduces the modeling and computation load. In this paper, if the minimum singular value of matrix $\boldsymbol{JE}$ is less than a certain value, it is judged that the system is in voltage instability. The voltage stability criterion of the 100% REPS is:

$$\inf_{\|\boldsymbol{x}\|=1} (\boldsymbol{JEx}) \geq \tau \tag{17}$$

where, $\boldsymbol{x}$ denotes a vector with L2 norm of 1.

While a straightforward approach to assessing voltage stability is to attempt solving the power flow equations at incrementally increased load levels, this method faces substantial challenges in inverter-dominated systems. The presence of continuous mode switching and strong nonlinearities can cause commercial solvers to either fail to converge prematurely or miss the true stability boundary. Moreover, such a brute-force approach is computationally intensive for large systems. The proposed Jacobian-free algorithm not only improves computational efficiency and robustness, but also enables continuous and accurate stability margin tracking, which is critical for real-time system monitoring and control in 100% renewable power grids.

C. Jacobian-Free Solution Method of Correction Equation Based on Krylov Subspace Iterative Algorithm

In the traditional power flow solution method, the Jacobian matrix is required to solve the correction equation. To avoid establishing the Jacobian matrix, it is necessary to establish a Jacobian-free correction equation solution method. Based on the Krylov subspace iterative algorithm, the Jacobian-free correction equation solution method is given below. The power flow equations can be abbreviated as:

$$\boldsymbol{f}(\boldsymbol{x}) = 0 \tag{18}$$

Given the initial value of variables, the correction equation can be obtained by linearizing the power flow equation $\boldsymbol{f}$:

$$\Delta \boldsymbol{x} = \boldsymbol{J}^{-1} \Delta \boldsymbol{c} \tag{19}$$

where, $\Delta \boldsymbol{c}$ denotes the correction error, $\Delta \boldsymbol{x}$ denotes the variable change vector to be solved.

In (19), the Jacobian matrix needs to be established and inverted, which will bring a huge computational burden in large-scale 100% REPSs. To avoid deriving the Jacobian matrix J, a Jacobian-free correction equation solution method based on the Krylov subspace iteration algorithm is proposed below. The Krylov subspace generated by the Jacobian matrix $\boldsymbol{J}$ and the correction error $\Delta \boldsymbol{c}$ is:

$$\kappa_m(\boldsymbol{J}, \Delta \boldsymbol{c}) = \text{span}\left\{\Delta \boldsymbol{c}, \boldsymbol{J}\Delta \boldsymbol{c}, \ldots, \boldsymbol{J}^{m-1}\Delta \boldsymbol{c}\right\}, m \leq n \tag{20}$$

where, κ_{m} denotes the Krylov subspace, span{•} denotes the expansion space, m denotes the dimension of the subspace.

The m column vectors in the subspace are linearly independent, that is, $\dim(\kappa_{\mathrm{m}}) = m$. Let $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_{\mathrm{m}}\}$ be a set of basis in the Krylov subspace κ_{m}, then any vector $\boldsymbol{v}$ in κ_{m} can be expressed as:

$$\boldsymbol{v} = \boldsymbol{W}_m \boldsymbol{y} \tag{21}$$

Algorithm 1 Jacobian-Free Arnoldi Process

1. $\boldsymbol{w}_1 = \Delta \boldsymbol{c} / \|\Delta \mathbf{c}\|_2$
2. for $j = 1$ to m do
3. $\mathrm{z} = \lim_{\varepsilon \to \mathbf{0}} \frac{\mathrm{f}(\mathrm{x}+\varepsilon \mathrm{w_j}) - \mathrm{f}(\mathrm{x})}{\varepsilon}$
4. for $i = 1$ to j do
5. $h_{i,j} =< \mathbf{w_i}, z >$
6. $z = z\text{-}h_{i,j}\mathbf{w_i}$
7. end for
8. $h_{j+1,i} = \|z\|_2$
9. if $h_{j+1,i} = 0$ then
10. break
11. end if
12. $\boldsymbol{w}_{j+1} = z/h_{j+1,i}$
13. end for

$$\begin{cases} \boldsymbol{W}_m = [\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_m] \\ \boldsymbol{y} = \left[\gamma_1, \gamma_2, \cdots, \gamma_m\right]^T \end{cases} \tag{22}$$

where, γ_i denotes linear expression coefficient.

Assuming that the best solution approximation in the Krylov subspace κ_{m} is $\Delta \boldsymbol{v}_{\mathrm{m}}$, the problem of finding $\Delta \boldsymbol{v}_{\mathrm{m}}$ can be solved by the following steps:

(1) Find a suitable set of basis $\{\mathbf{w_k}\}_{k=1}^{m}$;
(2) Solve the linear expression coefficient $\boldsymbol{y}$ of $\Delta \boldsymbol{v}_{\mathrm{m}}$.

First, the selection of basis is introduced. The vectors in the Krylov subspace are linearly independent, which can be used to form a set of basis. However, we hope to find a set of orthonormal basis to facilitate subsequent calculation and analysis. The following proposes a Jacobian-free Arnoldi process based on the modified Gram-Schmidt orthogonalization, which normalizes and orthogonalizes the vectors in the Krylov subspace.

The Jacobian-free Arnoldi process based on the modified Gram-Schmidt orthogonalization can be divided into the following four steps:

Step 1: Assuming that the first j-1 vectors have been normalized and orthogonalized, use the Jacobian-free method to obtain $\boldsymbol{J}\boldsymbol{w}_j$:

$$z = \boldsymbol{J}\boldsymbol{w}_{\mathbf{j}} = \lim_{\varepsilon \to \mathbf{0}} \frac{\boldsymbol{f}\left(\boldsymbol{x} + \varepsilon \boldsymbol{w}_{\mathbf{j}}\right) - \boldsymbol{f}(\boldsymbol{x})}{\varepsilon} \tag{23}$$

Step 2: Subtract the projection of the first j-1 vectors in z:

$$z = z - \sum_{i=1}^{j} \langle \boldsymbol{w}_i, z \rangle \boldsymbol{w}_i \tag{24}$$

where, $< \bullet, \bullet >$ denotes the inner product of two vectors.

Step 3: Assign a base vector $\boldsymbol{w}_j$ and normalize it:

$$\boldsymbol{w}_j = \frac{z}{\|z\|_2} \tag{25}$$

Step 4: Repeat the above process until all vectors are normalized and orthogonalized.

The detailed steps of the Jacobian-free Arnoldi process are shown in Algorithm 1.

By the Jacobian-free Arnoldi process, the base matrix $\boldsymbol{W}_{\mathrm{m}}$ will become an identity orthogonal matrix, and an upper

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Hessenberg matrix will be obtained. Suppose the upper Hessenberg matrix is:

$$\boldsymbol{H}_{m+1,m}=\begin{bmatrix} h_{1,1} & h_{1,2} & \cdots & h_{1,m} \\ h_{2,1} & h_{2,2} & \cdots & h_{2,m} \\ & h_{3,2} & \cdots & h_{3,m} \\ & & \ddots \ \ddots & \vdots \\ & & & h_{m+1,m} \end{bmatrix} \in \mathbf{R}^{(m+1)\times \mathbf{m}} \tag{26}$$

where, $\boldsymbol{H}_{m+1,m}$ denotes the upper Hessenberg matrix.

Denoting the base matrix after normalization and orthogonalization as $\boldsymbol{D}_{\mathrm{m}}$, the solution of the modified equation can be expressed as:

$$\Delta \boldsymbol{x}=\boldsymbol{D}_m \boldsymbol{y} \tag{27}$$

In (27), the coefficient vector $\boldsymbol{y}$ remains to be solved. A method for solving the coefficient vector $\boldsymbol{y}$ based on the least square method is given below. The solution of the coefficient vector $\boldsymbol{y}$ can be transformed into the following problem:

$$\begin{cases} \min \|\boldsymbol{r}\|_2 \\ \boldsymbol{r}=\Delta \boldsymbol{c}-\boldsymbol{J}\Delta \boldsymbol{x}=\Delta \boldsymbol{c}-\boldsymbol{J}\boldsymbol{W}_m \boldsymbol{y} \end{cases} \tag{28}$$

When the minimum L2 norm of the vector $\boldsymbol{r}$ reaches 0, the coefficient vector $\boldsymbol{y}$ is solved.

By the Jacobian-free method, we can get:

$$\boldsymbol{J}\boldsymbol{w}_j=\lim_{\varepsilon \to 0}\frac{\boldsymbol{f}\left(\boldsymbol{x}+\varepsilon \boldsymbol{w}_j\right)-\boldsymbol{f}\left(\boldsymbol{x}\right)}{\varepsilon} \tag{29}$$

According to the Jacobian-free Arnoldi process, (24) can be rewritten as:

$$h_{j+1,j}\boldsymbol{w}_{j+1}=\lim_{\varepsilon \to 0}\frac{\boldsymbol{f}\left(\boldsymbol{x}+\varepsilon \boldsymbol{w}_j\right)-\boldsymbol{f}\left(\boldsymbol{x}\right)}{\varepsilon}-\sum_{i=1}^{j} h_{i,j}\boldsymbol{w}_i \tag{30}$$

Combining (29) and (30), we get:

$$\begin{aligned} \boldsymbol{J}\boldsymbol{w}_j &= \boldsymbol{W}_{m+1}\left[h_{1,j},\cdots,h_{j+1,j},0,\cdots,0\right]^T \\ &= \boldsymbol{W}_{m+1}\boldsymbol{H}_{m+1,m}\left(:,j\right) \end{aligned} \tag{31}$$

According to (31), the product of the Jacobian matrix $\boldsymbol{J}$ and the Krylov subspace basis matrix $\boldsymbol{W}_{\mathrm{m}}$ can be expressed as:

$$\boldsymbol{J}\boldsymbol{W}_m=\boldsymbol{W}_{m+1}\boldsymbol{H}_{m+1,m} \tag{32}$$

Eq. (28) can be rewritten as:

$$\boldsymbol{r}=\Delta \boldsymbol{c}-\boldsymbol{W}_{m+1}\boldsymbol{H}_{m+1,m}\boldsymbol{y} \tag{33}$$

The $\boldsymbol{w}_1$ satisfy:

$$\boldsymbol{w}_1=\frac{\Delta \boldsymbol{c}}{\|\Delta \boldsymbol{c}\|_2} \tag{34}$$

Therefore, Eq. (33) can be expanded as:

$$\begin{aligned} \boldsymbol{r} &= \|\Delta \boldsymbol{c}\|_2 \boldsymbol{w}_1-\boldsymbol{W}_{m+1}\boldsymbol{H}_{m+1,m}\boldsymbol{y} \\ &= \boldsymbol{W}_{m+1}\left(\|\Delta \boldsymbol{c}\|_2 \boldsymbol{e}_1-\boldsymbol{H}_{m+1,m}\boldsymbol{y}\right) \end{aligned} \tag{35}$$

where, $\boldsymbol{e}_1$ denotes a column vector of length $m+1$, only the first element is 1, and the rest is 0, therefore $\boldsymbol{W}_{\mathrm{m}+1}\mathbf{e}_1$ is $\boldsymbol{w}_1$.

Set $\mu=\|\Delta \boldsymbol{c}\|_2$, then the L2 norm of the error vector $\boldsymbol{r}$ can be written as:

$$\|\boldsymbol{r}\|_2=\left\|\boldsymbol{W}_{m+1}\left(\mu \boldsymbol{e}_1-\boldsymbol{H}_{m+1,m}\boldsymbol{y}\right)\right\|_2=\left\|\mu \boldsymbol{e}_1-\boldsymbol{H}_{m+1,m}\boldsymbol{y}\right\|_2 \tag{36}$$

Algorithm 2 Jacobian-free correction equation solution algorithm

1. Set the termination gap ε, the maximum number of iterations IterMax
2. for m = 1 to IterMax do
3. Generate the identity orthogonalization basis $\boldsymbol{W}_{\mathrm{m}}$ and upper Hessenberg matrix $\boldsymbol{H}_{m+1,m}$ by Arnoldi process
4. Establish the least squares problem and solve it to obtain the coefficient vector $\boldsymbol{y}$
5. Get the residual= $\|\boldsymbol{r}\|_2$
6. if residual is smaller than ε then
7. break
8. end if
9. end for
10. Solve the correction equation $\Delta \boldsymbol{v}=\boldsymbol{W}_{\mathrm{m}}\boldsymbol{y}$

After selecting the dimension m and the vector base $\boldsymbol{W}_{\mathrm{m}}$, the coefficient vector $\boldsymbol{y}$ should satisfy:

$$\boldsymbol{y}=\underset{\mathbf{y}\in \mathbb{R}^{\mathbf{m}}}{\arg\min}\left\|\mu \boldsymbol{e}_1-\boldsymbol{H}_{m+1,m}\boldsymbol{y}\right\|_2 \tag{37}$$

Eq. (37) is a least squares problem. Since $H_{m+1,m}$ is an upper Hessenberg matrix, Eq. (37) can be solved by QR decomposition based on Givens transformation. For the selection of m, it should be as small as possible while meeting the accuracy requirements to improve computational efficiency. The Jacobian-free correction equation solution algorithm can be summarized as Algorithm 2.

D. SVS Estimation

Utilizing the proposed Jacobian-free voltage stability criterion and the correction equation solution method, a SVS estimation method is given below. The parameterized power flow equations of the system can be abbreviated as:

$$\boldsymbol{f}\left(\boldsymbol{x},\lambda\right)=\mathbf{0} \tag{38}$$

where, λ denotes the load increasing coefficient.

Under the action of λ, the load and RES output increase simultaneously, as (39)–(40) show:

$$\begin{cases} P_{\mathrm{L}i}\left(\lambda\right)=\left(1+\lambda\right)P_{\mathrm{L}i0} \\ Q_{\mathrm{L}i}\left(\lambda\right)=\left(1+\lambda\right)Q_{\mathrm{L}i0} \end{cases} \tag{39}$$

$$P_{Gi}=P_{Gi0}\left(1+\lambda\right) \tag{40}$$

where, $P_{\mathrm{L}i}$ and $Q_{\mathrm{L}i}$ denote the active and reactive power of the load respectively, $P_{\mathrm{L}i0}$ and $Q_{\mathrm{L}i0}$ denote the initial active and reactive power of the load respectively, $P_{\mathrm{G}i}$ denotes the output power of RESi, $P_{\mathrm{G}i0}$ denotes the initial output power of RESi.

The purpose of the SVS estimation method is to find a critical value of λ at which the system is in a critical voltage instability. The method can be divided into three parts, including prediction, correction and judgment.

1) Prediction: The purpose of prediction is to provide initial values for the correction. Assuming that a steady-state

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Algorithm 3 Solution method of augmented power flow equations based on JFKA

1. Set the initial values of $\boldsymbol{x}$ and λ, the convergence criterion ε, and the maximum number of iterations IterMax
2. for $j = 1$ to IterMax do
2. Calculate power error $g(\boldsymbol{x}_j, \lambda) = \Delta \boldsymbol{c}$
3. Get $\Delta \boldsymbol{x}$ according to the proposed Jacobian-free correction equation solution algorithm
4. Correct $\boldsymbol{x}_{j+1} = \mathbf{x_j}\text{-}\Delta \boldsymbol{x}$
5. if $\| g(\boldsymbol{x}_{j+1})\|_2 < \varepsilon$ then
6. break
7. end if
8. End for

operating point ($\boldsymbol{x}_0$, λ_0) is given, at this operating point, Eq. (38) can be differentiated on both sides at the same time:

$$d\boldsymbol{f} = \frac{d\boldsymbol{f}}{d\boldsymbol{x}} d\boldsymbol{x} + \frac{d\boldsymbol{f}}{d\lambda} d\lambda = 0 \tag{41}$$

Due to the introduction of the load increasing coefficient λ, the number of the above equations is 1 less than the number of unknowns, and a new equation needs to be added. The specific method is to specify the variation of λ, so that (41) becomes:

$$\begin{bmatrix} \frac{d\boldsymbol{f}}{d\boldsymbol{x}} & \frac{d\boldsymbol{f}}{d\lambda} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} d\boldsymbol{x} \\ d\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ \mu \end{bmatrix} \tag{42}$$

where, μ denotes the variation of λ.

Using the Jacobian-free correction equation solution method to solve (42), then predict the solution of (38) according to the tangent vector:

$$\begin{bmatrix} \tilde{\boldsymbol{x}}^{(i+1)} \\ \tilde{\lambda}^{(i+1)} \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}^{(i)} \\ \lambda^{(i)} \end{bmatrix} + \begin{bmatrix} d\boldsymbol{x} \\ d\lambda \end{bmatrix} \tag{43}$$

where, $\tilde{\boldsymbol{x}}^{(\mathbf{i+1})}$ and $\tilde{\lambda}^{(i+1)}$ denote the estimated values of $\boldsymbol{x}$ and λ for the $i+1$-th iteration respectively, $\boldsymbol{x}_i$ and λ_i denote the real values of $\boldsymbol{x}$ and λ for the i-th iteration respectively.

2) Correction: After obtaining the predicted value of x and λ, it can be used as the initial value of the power flow equation. Create the augmented power flow equation:

$$\boldsymbol{g}(\boldsymbol{x}, \lambda) = \begin{bmatrix} \mathrm{f}(\boldsymbol{x}, \lambda) \\ \lambda - \tilde{\lambda} \end{bmatrix} = \mathbf{0} \tag{44}$$

For the solution of the augmented power flow equations $\boldsymbol{g}(\boldsymbol{x}, \lambda)$, a JFKA solution method is given, and the detailed steps are shown in Algorithm 3.

3) Judgement: Use the proposed Jacobian-free voltage stability criterion to judge whether the load limit has been reached. If the load limit is not reached, return to the first step to increase the load increasing coefficient λ to predict and correct again. If the load limit is reached, then the estimation for the load margin is completed.

IV. Case Study

The effectiveness and feasibility of the proposed method is verified in the modified IEEE 39 bus system. For the IEEE 39 bus system, the original ten SGs are replaced by ten RESs. The outputs of the RESs are consistent with those of original SGs, maintaining the power flow unchanged. The topology of the modified IEEE 39 bus system is shown in Fig. 4, and the main parameters of the RESs are shown in Table. II.

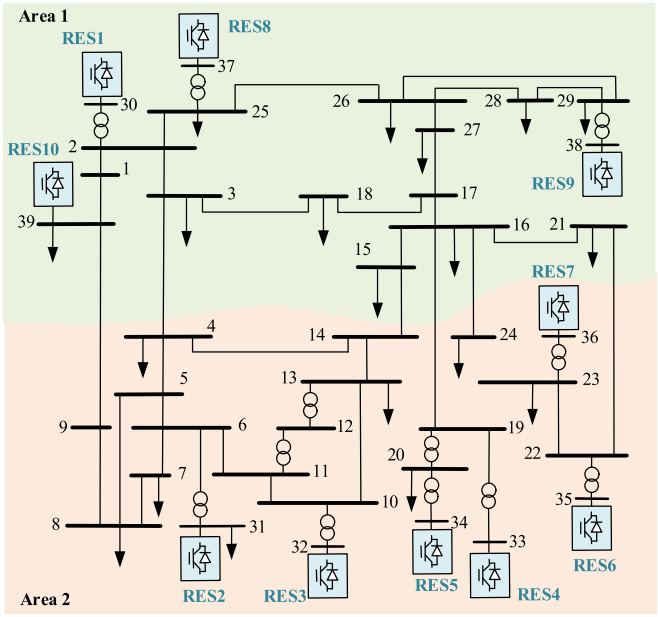


Fig. 4. Topology of the modified IEEE 39-bus system.

TABLE II
Main Parameters of the RESs

	Output power (pu)	Rated capacity (pu)	Maximum allowable output current (pu)
RES1	2.5	2.8	3.4
RES2	6.78	7.2	8.6
RES3	6.5	7.0	8.4
RES4	6.32	6.5	7.8
RES5	5.08	5.3	6.4
RES6	6.5	6.5	7.8
RES7	5.6	5.6	6.7
RES8	5.4	5.4	6.5
RES9	8.3	8.3	10.0
RES10	10.0	20.0	24.0

A. Verification of Jacobian-Free Correction Equation Solution Method

Firstly, the accuracy of the proposed Jacobian-free correction equation solution method is verified. For the calculation result $\boldsymbol{x}_\mathrm{r}$ of the power flow equation $\boldsymbol{f}(\boldsymbol{x}) = 0$, the following equation can be used to evaluate the accuracy:

$$\beta = \|\mathbf{f}(\mathbf{x_r})\|_2 \tag{45}$$

where β is the L2 norm of the residual error vector.

The dimension of the Krylov subspace is given by the user, which has a great influence on the accuracy of the proposed method. To this end, the accuracy of the proposed method is analyzed under different Krylov subspace dimensions. The modified IEEE 39 bus system contains 29 PQ buses, and each PQ bus is described by (6), with a total of 58 power balance equations; 9 PV buses, each is described by (7), (8), and (10), with a total of 54 current or power equations; 1 balance bus,

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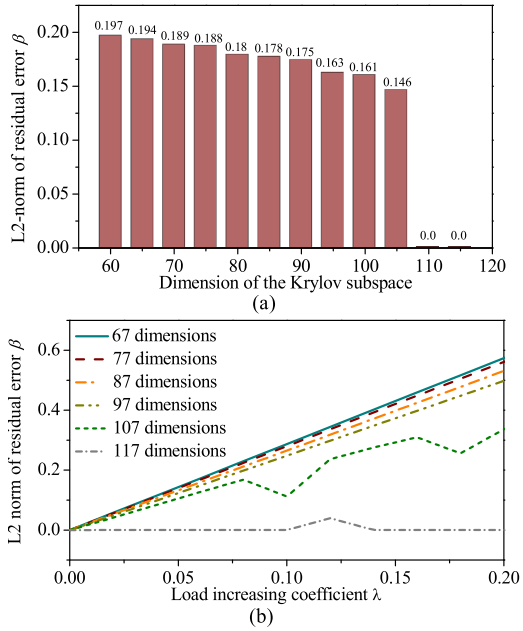


Fig. 5. Verification of Jacobian-free correction equation solution method. (a) The L2 norm β versus the Krylov subspace dimension. (b) The L2 norm β versus the load increasing coefficient.

described by (7), (8), and (11), with a total of 5 current or power equations. Therefore, for the modified IEEE 39 bus system, the order of its generalized power flow model is 117, which is also the maximum dimension of the Krylov subspace. When the dimension is selected as 117, the advantage of the Krylov subspace in improving computational efficiency no longer exists. On the premise of meeting accuracy, the smaller the dimension of the Krylov subspace, the better the computational efficiency of the proposed method.

Take the load increasing coefficient λ as 0.1, and chang the dimension of the Krylov subspace, the L2 norm β of the residual error vector versus the dimension is shown in Fig. 5(a). It can be seen from Fig. 5(a) that as the dimension of the Krylov subspace increases, the L2 norm β gradually decreases. When the dimension is less than 105, the decreasing rate is low. When the Krylov subspace dimension reaches 110, the L2 norm β drops off a cliff, which shows that the proposed method can solve the power flow equation almost without error. However, when the dimension of the Krylov subspace reaches 110, the advantage of computational efficiency of the proposed method is no longer prominent.

In addition to the Krylov subspace dimension, the load increasing coefficient λ also has a great impact on the accuracy. The smaller the load increasing coefficient λ, the closer the predicted value is to the real value, and the real solution can be obtained using the Krylov subspace with fewer dimensions. The L2 norm variation of the residual error vector with the load increasing coefficient is shown in Fig. 5(b). As per Fig. 5(b), with the increase of the load increasing coefficient, the performance difference of the proposed method with different dimensions will gradually increase. When the load increasing coefficient does not exceed 0.02, the L2 norm β does not exceed 0.1. Therefore, when applying the proposed

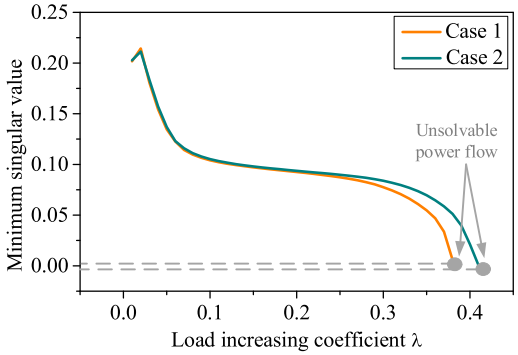


Fig. 6. The minimum singular value of matrix (15) versus the load increasing coefficient.

method, to make a tade-off between computational efficiency and accuracy, a feasible approach is to choose a small Krylov subspace dimension and a small load increasing coefficient.

It should be clarified that the dimension of the Krylov subspace is a user-defined parameter for the iterative solution and is not limited by the system size. The number 117, in this case, corresponds to the order of the generalized power flow equations for the modified IEEE-39 bus system and is used solely for demonstration purposes. For practical large-scale power systems, the Krylov subspace dimension can be chosen much smaller than the system order to balance computational efficiency and solution accuracy. As a result, the proposed JFKA framework remains fully scalable and suitable for real-world systems of much larger size.

B. Verification of The Jacobian-Free Voltage Stability Criterion

Generally speaking, the voltage instability is equivalent to the insolvability of the power flow equations, another manifestation is the singularity of the Jacobian matrix. In this paper, a Jacobian-free voltage stability criterion is proposed. The proposed criterion avoids the establishment of the Jacobian matrix, and estimates the voltage stability by judging the singularity of matrix (15), which can be obtained by (16). To verify the effectiveness of the Jacobian-free voltage stability criterion, the relationship between the singularity of matrix (15) and the voltage stability is studied below.

The variation of the minimum singular value of matrix (15) with the load increasing coefficient in two cases are shown in Fig. 6. In Case 1, only the active load increases with the load increasing coefficient, while the reactive load remains unchanged. In Case 2, both the active load and reactive load increase with the load increasing coefficient. It can be seen from Fig. 6 that with the load increasing, the minimum singular value of matrix (15) decreases continuously. Then we can conclude that the Jacobian-free voltage stability criterion can accurately evaluate the voltage stability of the system. As per Fig. 6, when the power flow equation has no solution, the minimum singular value of matrix (15) is at a very small value. Therefore, the effectiveness of the proposed Jacobian-free voltage stability criterion is verified.

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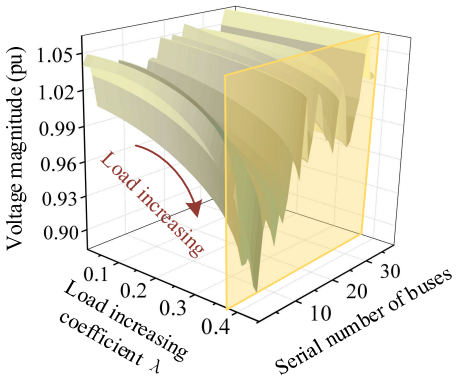


Fig. 7. Voltage magnitude of each bus versus load increasing coefficient.

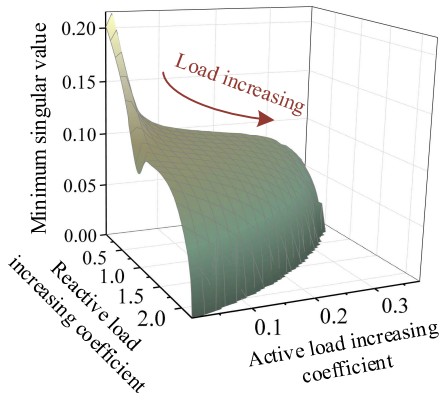


Fig. 8. Minimum singular value versus active and reactive load.

Assuming that the active and reactive load increase in proportion to the load increasing coefficient, the voltage magnitude of each bus versus the load increasing coefficient is shown in Fig. 7. As per Fig. 7, with the load increasing, the voltage magnitude of each bus decreases. However, the reduction size varies among different buses, with a smaller decrease in the voltage magnitude of the buses near power sources, while a larger decrease in the voltage magnitude of buses far from the power sources. Among all buses, bus 8 has the largest decrease. When the load increasing coefficient increases to 0.38, the voltage magnitude of bus 8 decreases from 1.0 pu to 0.88 pu. When the load increasing coefficient reaches 0.38, according to the Jocabian-free voltage stability criterion, the system is in a critical voltage instability. Therefore, for the modified IEEE 39-bus system, the SVS estimated by the load increasing coefficient is 0.38.

C. Impact of Load Increasing Direction

As mentioned above, the maximum load increasing coefficient of the modified IEEE 39 bus system is 0.38. This value is obtained by increasing all loads in equal proportion, the active and reactive power of each load also increases proportionally. Due to the spatial distribution of voltage, the load increasing direction also has a great influence on the SVS. With different load increasing directions, the SVS will also be different. In the following, the impact of load increasing direction on the SVS is studied.

The variation of the minimum singular value with the increasing coefficient of active and reactive loads is shown in Fig. 8. In Fig. 8, all active or reactive loads increase proportionally. From Fig. 8, it can be seen that for the modified IEEE-39 bus system, the main reason for voltage instability is the growth of active load, since the power factor of the load is generally large. In addition, as the reactive load increases, the allowable active load growth decreases. Therefore, compensating all reactive loads on site is of great significance for improving the SVS.

The previous analysis is based on the proportional increase of all active or reactive loads. To investigate which load has the greatest impact on the SVS of the system, the load limit that the system can withstand is studied for single load

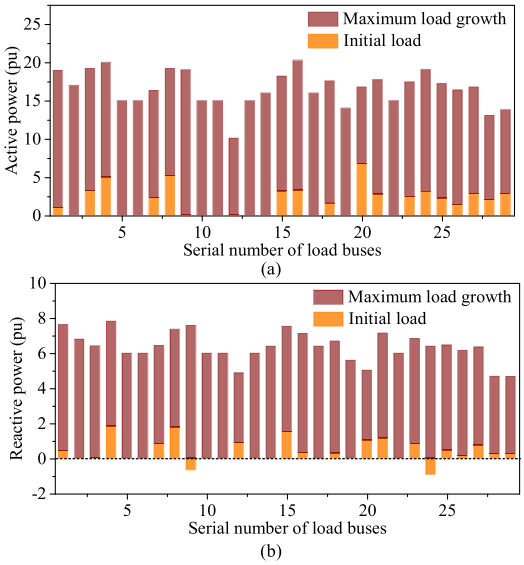


Fig. 9. The load limit of each bus can withstand. (a) Active load limit of each bus can withstand. (b) Reactive load limit of each bus can withstand.

increasing scenarios. For the 29 PQ buses, gradually increase the connected load, and obtain the maximum active or reactive load limit that each bus can withstand, as shown in Fig. 9. As per Fig. 9, the active load limit that bus 12 can withstand is the smallest, only 9.8 pu, while the active load limit that bus 16 can withstand is the largest, reaching 20.3 pu. The reactive load limit that each bus can withstand is shown in Fig. 9(b). According to Fig. 9(b), the reactive load limit is generally lower than the active load limit. The reason is that the relationship between voltage magnitude and reactive power is closer, the transmission of reactive power is more likely to cause voltage drop.

D. Impact of RES Mode Switching

Compared with the traditional power system dominated by SGs, the biggest difference in the SVS of 100% REPSs is that the influence of the RES mode switching needs to be taken into account. When the bus voltage is at a low level, the RES will switch from constant voltage control mode to constant current control mode, causing SVS deterioration. To

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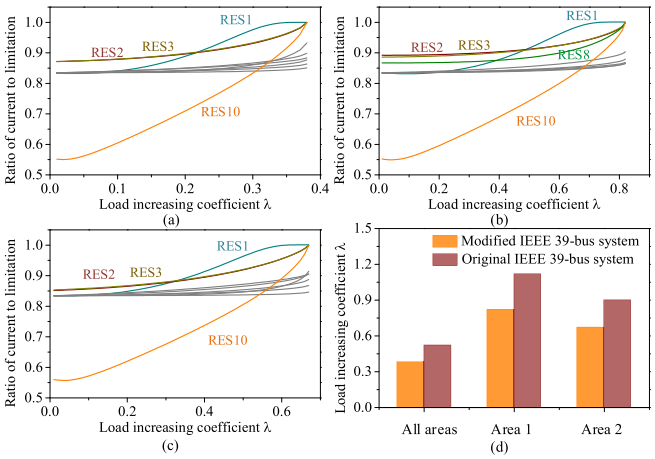


Fig. 10. Impact analysis of the RES current limiting. (a) Output current of each RES when all loads increase in equal proportion. (b) Output current of each RES when loads in Area1 increase. (c) Output current of each RES when loads in Area2 increase. (d) The maximum load increasing coefficient of the IEEE 39 bus system before and after modification.

analyze the impact of control mode switching, we calculate the current change of each RES with the load increasing coefficient, the calculations are carried out in three cases with different load increasing directions. When all loads increase in equal proportion, the current variation of each RES is shown in Fig. 10(a). In Fig. 10(a), the abscissa is the load increasing coefficient, and the ordinate is the ratio of the RES current to its own limit. As per Fig. 10(a), as the load increases, the current of RES1 first reaches the limit, the load increasing coefficient is 0.33 at this time. The system does not experience voltage instability until the load increasing coefficient reaches 0.38. When the output currents of RES1, RES2, RES3 and RES10 reach the limit at the same time, the system becomes unstable. The voltage instability is caused by the current limits of four RESs (RES1, RES2, RES3, and RES10). When only the loads in Area 1 increase, the current variation of each RES is shown in Fig. 10(b), and the load increasing coefficient can reach 0.82 at most. At the critical stable point, the output currents of the five RESs of RES1, RES2, RES3, RES8 and RES10 reach the limit. The current of each RES versus the load growth in Area 2 is shown in Fig. 10(c). The load increasing coefficient can reach a maximum of 0.63. When the static voltage instability occurs, RES1, RES2, RES3 and RES10 reach the current limit at the same time.

To illustrate the influence of the RES current limit on the SVS, the calculation is carried out in the modified IEEE 39-bus system with 100% renewable energy generation and the original IEEE 39-bus system respectively. The maximum load increasing coefficient that the system can withstand is obtained as Fig. 10(d) shows. It can be seen from Fig. 10(d) that when all loads increase proportionally, the modified IEEE 39-bus system can withstand a maximum load increasing coefficient of 0.38, while the original IEEE 39-bus system can withstand a maximum load increasing coefficient of 0.52. Similarly, if the loads in Area1 or Area2 increase independently, the maximum load increasing coefficient that the original IEEE 39-bus system can withstand is much larger than that of the

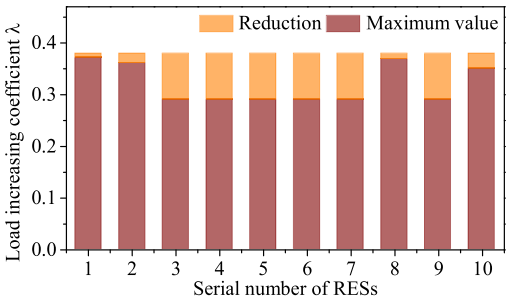


Fig. 11. Analysis on the influence of each RES on the SVS.

TABLE III
MAIN PARAMETERS OF THE RESS

Method	Computation Time /s	Converged Margin λ	Did Converge?
CPF [11]	15.2	0.37	Yes
Holomorphic Embedding [12]	30.7	0.36	Partial
load-impedance modulus margin [13]	22.3	0.45	Yes
Static Inverter Model [23]	6.8	0.42	Yes
Proposed JFKA	2.7	0.38	Yes

modified IEEE 39-bus system. This indicates that due to the effect of RES current limiting, the SVS of the 100% REPS is significantly reduced.

According to the analysis results in Fig. 10, it can be seen that the current limit of the RES has a great influence on the SVS of the system. To investigate the influence of each RES on the SVS, the current limit of each RES is modified one by one, and then the maximum load increasing coefficient that the system can withstand is calculated. The specific steps are to select a RES and reduce its current limit to 0.8 times the initial value, then increase the load of the entire system proportionally, calculate the maximum load increasing coefficient that the system can withstand, and compare it with the original result. The comparison results are shown in Fig. 11. It can be seen from Fig. 11 that the influence of the current limit of RES3, RES4, RES5, RES6, RES7 and RES9 on the SVS is basically the same, when the current limit of any one of these RESs is reduced to 0.8 times of the initial value, the maximum load increasing coefficient that the system can withstand will be reduced from 0.38 to 0.29.

E. Comparison Between Different Methods

As shown in Table III, the proposed JFKA method exhibits distinct advantages over representative existing approaches. While CPF can generally converge to a reasonable voltage stability margin, it requires significantly longer computation time due to repeated Jacobian evaluations and rollbacks near the bifurcation point. Holomorphic embedding shows convergence issues as the system approaches collapse, resulting in longer computation time and lower accuracy near the boundary. The static inverter model, which ignores dynamic mode-switching and current-limiting effects, gives an overestimated margin

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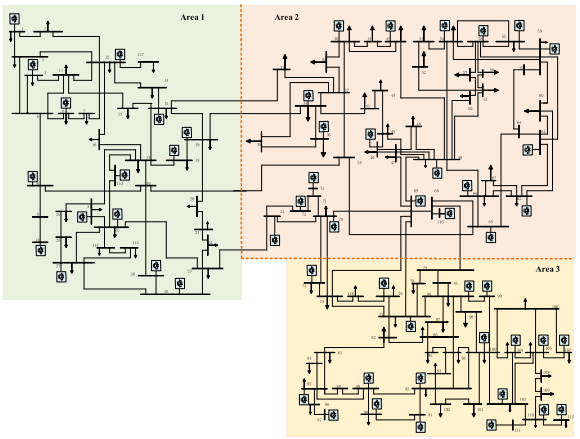


Fig. 12. Analysis on the influence of each RES on the SVS.

and may misinform operators. In contrast, the proposed JFKA framework achieves robust convergence, accurate margin estimation, and the shortest computation time, demonstrating its suitability for large-scale, real-time assessment in 100% renewable systems.

F. Verification in IEEE 118 Bus System

To verify the proposed method in a larger-scale power system, the IEEE 118 bus system is modified to be a power system with purely renewable energy generation. Compared with the original IEEE 118 bus system, the generators are replaced by RESs, and the condensers are replaced by STATCOMs. After modification, the system contains 19 RESs and 35 STATCOMs, 177 lines, 9 transformers, and 91 loads. The output of RES is consistent with that of the original RES, ensuring that the power flow of the whole system remains unchanged. The topology of the modified IEEE 118 bus system as shown in Fig. 12.

When all SGs are replaced by RES, the current limit of RES is set to 1.2 times the rated capacity, while the rated capacity of RES remains the same as the original SGs. Table II shows the evaluation results of SVS before and after system modification. In addition, Table II also shows the SVSME results when reducing the current limit of each RES to 0.8 times the original value. It can be seen from Table II that for the original IEEE 118 bus system, the maximum load increasing coefficient it can withstand is 0.54, but after replacing all SGs with RESs, the maximum load increasing coefficient is reduced to 0.42. With the reduction of the current limit of RES, the maximum load increasing coefficient will be significantly reduced. When the current limit of each RES is reduced to 0.8 times of the initial value, the maximum load increasing coefficient is only 0.11. When the system is in critical static voltage instability, the lowest voltages in the three cases all occur on bus 53. For the modified IEEE 118 bus system with 100% renewable energy generation, the RESs that reach the current limit in critical static voltage instability are also shown in Table II. Compared with the original system, the SVS of the modified system deteriorates, and the RES that reaches the current limit is the cause of the SVS deterioration.

TABLE IV
ANALYSIS OF IMPACT OF THE RES CURRENT LIMIT ON THE SVS

Items	0.8 times the current limit	Initial current limit	Original system composed entirely of SGs
Maximum load increasing coefficient	0.11	0.42	0.54
Lowest bus voltage	0.94	0.93	0.92
Bus with lowest voltage	53	53	53
RESs reaching the current limit	10, 65, 80, 89, 69	49, 80, 89, 90, 69	-

TABLE V
ANALYSIS OF IMPACT OF RESS IN DIFFERENT AREAS ON SVS

Items	Area1	Area2	Area3
Maximum load increasing coefficient	0.52	0.5	0.48
Lowest bus voltage	0.92	0.93	0.93
Bus with lowest voltage	53	53	53
RESs reaching the current limit	8, 10	49, 65, 69	80, 89, 90

The modified IEEE 118 bus system can be divided into three areas, as shown in Fig. 12. On the basis of the original IEEE 118 bus system, replace SGs in one area with RESs, and then estimate the SVS margin of the system. The results are shown in Table V. When the SG in Area 1 is replaced by RES, area 2 and area 3 remain unchanged. In this case, the maximum load increasing coefficient that the system can withstand is 0.52. Compared with the original system composed entirely of SGs, the maximum load increasing coefficient that the system can withstand is slightly reduced. When the load increasing coefficient reaches 0.52, the output currents of the RESs connected to buses 8 and 10 reach the limit, which is the cause of the decrease in the SVS. If the capacity of RESs connected to buses 8 and 10 is increased, the SVS of the system can be effectively improved. Similarly, if the SG in Area 2 or Area 3 is replaced by RES, the RESs that limit the load increasing are also shown in Table III. In addition, after the SG is replaced by the RES in any area, the lowest voltage occurs on bus 53.

V. CONCLUSION

This paper presents a unified JFKA framework for SVS estimation in 100% REPSs, explicitly considering the control mode switching behavior of RESs. By introducing a smooth S-type function to model inverter current limiting and developing a singularity-based, Jacobian-free voltage stability criterion, the proposed method effectively addresses the nonlinear and mode-dependent challenges inherent to 100% RESPs. The integration of a reduced-order Arnoldi process within a Newton-Krylov solver enables efficient and memory-saving solutions for large-scale, transcendental power flow equations. Comprehensive case studies on a modified IEEE-39 bus system demonstrate that the JFKA framework accurately tracks SVS under various load growth scenarios and control mode transitions. Results confirm that inverter current limiting

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and mode switching significantly reduce the SVS compared to traditional systems.

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